Mega cheat sheet: Exam 1 (copy by hand, front and back)

Before you copy: ask Prof. Smith whether AI-assisted notes may go on the sheet (see the caution in `cheat-sheet.md`). Copying by hand is the point either way: rework one problem per section from the sheet alone as you go.

Layout: FRONT = the 65 computation points. BACK = planes, leverage, words, rounding, clock. Section numbers match blueprint items. Anchors (in brackets) are worked answers to check yourself.

FRONT

A. Home equation (everything) [items 3, 7-11]

```
Rev − VC − FC = Profit
P(V) − vc(V) − FC = Profit
CM/u(V) − FC = Profit
Rev − VC%(Rev) − FC = Profit
TC = vc(V) + FC          CM ratio = 1 − VC ratio
```

Write it, fill 3, solve the 4th. NEVER unitize FC or profit: x back to totals first.

B. Single-product CVP [7-9, 15 pts]

```
BE units = FC / CM/u                ROUND UP
BE $     = FC / CM ratio
Target V = (FC + Target) / CM/u    ROUND UP
Pretax   = After-tax / (1 − t)
           DIVIDE, never x(1−t)
```

Return target: (1) % x base = after-tax $ (2) / (1-t) (3) into home eq. ROE -> EQUITY. ROA -> TOTAL ASSETS. Rest of balance sheet = decoy. Target as % of revenue: [CM/u - k(P)]V = FC (can't add it to FC).

[P 8, vc 4.20, FC 180k: BE 47,369 u, $378,947; +120k target -> 78,948]

[500k after tax @30% -> 714,286 pretax. tax-003: 30% ROE on 300k equity, t 40% -> 150k pretax -> P 50.80]

C. Margin of safety [7-9]

```
MoS = Current − BE   (units or $)
"how far ABOVE BE?"        MoS / BE
   [12,631/47,369 = 26.7%]
"how far can sales FALL?"  MoS / CURRENT
   [12,631/60,000 = 21.05%]
```

The words pick the denominator.

D. Multi-product CVP [10-11, 15 pts]

Method	Weight by	Answer in
Basket	unit ratio	units per product
Wtd avg CM	UNIT share	total units
Revenue	REVENUE share	dollars

Basket: ratio -> basket P, basket VC -> basket CM -> baskets = (FC + target)/basket CM -> UNPACK, then round EACH product UP. Do not round baskets ("not a thing"). Revenue: BE $ = (FC + target) / (1 - total VC / total Rev). Split $ by REVENUE share. No unique multi-product BE: new mix = recompute from scratch.

[mock-1-10: 8:3, basket CM 480, 2,177.08 baskets -> 17,417 + 6,532]

[Goldman 3:1 -> 160,000; 9:1 -> 181,819. One-unit-off option = rounding order]

E. Cost behavior [2, 4, 16, 15 pts]

GAAP (COGS vs S&A) is NOT variable vs fixed. Rebuild: vc/u x V = total VC; total cost - VC = FC.

[mock-1-04: vc 35 x 24,000 = 840k; 1,292k - 840k = FC 452k]

```
High−low (pick by ACTIVITY, not cost):
  b = (Hi$ − Lo$) / (Hi act − Lo act)
  a = Hi$ − b(Hi act)
  CHECK a at the low point too
```

[mock-1-02: b 19.00, a 106,400 both points]

Step cost: resources = V / capacity, ROUND UP, then total. Never average to a per-unit rate.

[mock-1-16: 1,640/500 -> 4 techs; 29,600 + 46,000 + 31,160 = 106,760]

Relevant range: model only valid inside observed data. High-low = 2 points; regression = all.

F. Regression [5-6, 10 pts]

CI around	Use SE of
Fixed cost	INTERCEPT row
Variable cost	SLOPE (driver) row
Total cost estimate	REGRESSION (top block)

```
CI = point +/− t(SE)      df = n − 2
t @ 6 df: 70% 1.134 | 80% 1.440
          90% 1.943 | 95% 2.447
t stat = coef / own SE
  >=2 useful, >3 near certain. p < .05
Report ADJ R^2.
Point estimate from PRINTED rounded coefs,
  compute it FIRST.
```

[Sailmaster: 95% total @325 = 273,116.04 +/- 2.447(27,771.27) -> 205,159.74 to 341,072.34]

BACK

G. Planes: cost accounting [14-15, 10 pts]

Mfg costs (DM, DL, VOH, AND FIXED OH) attach to units -> inventory until sold. S&A = period cost, expensed now.

```
Units of prod, DM   = sold + ALL WIP
  (materials in at start)
Units of prod, conv = sold + WIP x % complete
Rate = cost / its own denominator
  (two denominators, never one)
Op profit = revenue - (rates x SOLD units)
End inv = DM rate x WIP + conv rates x (WIP x %)
TIE OUT: expensed + inventory = total spent
```

[23 started, 18 sold, 5 WIP @60%: denominators 23 and 21; profit 262.72; inv 51.21; 277.29 + 51.21 = 328.50]

Absorption: produce more -> FOH moves to balance sheet -> reported profit rises with no extra sales. Same as "equivalent units" / "work done".

H. DOL [12, 5 pts]

```
DOL = TOTAL CM $ / Op profit
  (not CM ratio, not revenue)
%chg OP = DOL x %chg Rev
  cuts BOTH ways; point-in-time,
  recompute after V moves
```

[mock-1-12: 836k / 308k = 2.71; +/-10% rev -> +/-27.1% OP]

I. Operating leverage, short answer [13, 5 pts]

OL = cost STRUCTURE (share of cost that is fixed), stable. DOL = sensitivity AT today's volume. High OL: more FC, higher CM/u, higher BE, smaller MoS, bigger swings both ways. Fixed-for-variable swap (machines for labor, buy vs royalty) RAISES OL and BE even as unit cost falls.

[Alpine vs Basin: same V, Basin DOL 2.81 vs 1.71; after -20% rev 206k vs 276k]

J. CVP short answer [3, 5 pts], one line each

- Unitized FC is true only at the volume where computed; CVP always moves volume.
- GAAP splits core vs run-the-business; COGS and S&A are both mixed.
- CM = revenue less ALL variable costs, wherever they sit.
- CVP uses: pricing, fixed-for-variable swaps, outsourcing, target profit. NOT quality specs.
- Two MoS %: denominator follows the question.

K. Case answers [17: 50 words, 18: 90 words, 20 pts]

Read the question twice. Answer THAT question. No preamble, no restating.

- Case 1 (cost structure): labor as variable = low BE, slow profit growth; as fixed = high BE, fast growth. Same profit today, different risk. Format: rec -> 2 driving numbers -> 1 caveat.
- Case 2 (shared cost, lines): allocation changes reported line profit ONLY, not total profit or cash. Basis = what causes the cost. Judge a line on CM - its AVOIDABLE FC. Dropping a contributing line leaves shared cost behind -> total profit falls. Finish on the recommendation.
- Hammer line: 12:1 unit mix ~ 2:1 revenue mix; BE $54,731 x 33% = 18,061 (exact 32.62% = 17,853).

L. Rounding

BE / required volume UP. Baskets never. Unpack then round each UP. Regression from printed coefs. He never takes points for rounding: pick the nearest option and move on.

M. Clock and Excel

2h15m, 18 items. ~6 min per computation; keep ~30 min for items 3, 13, 17, 18. Do the cases before the last 10 minutes. Excel starts blank. No GoalSeek, no Solver. Item 1 (integrity): read it, answer yes.